

For a more real-world example, the following are iterative and recursive functions that return the index of a substring, `needle`, in a string, `haystack`. The functions return `-1` if the substring isn't found. This is similar to Python's `find()` string method and JavaScript's `indexOf()` string method. This *findSubstring.py* program has a Python version:

Python

```
def findSubstringIterative(needle, haystack):
    i = 0
    while i < len(haystack):
        if haystack[i:i + len(needle)] == needle:
            return i # Needle found.
        i = i + 1
    return -1 # Needle not found.

def findSubstringRecursive(needle, haystack, i=0):
    if i >= len(haystack):
        return -1 # BASE CASE (Needle not found.)

    if haystack[i:i + len(needle)] == needle:
        return i # BASE CASE (Needle found.)
    else:
        # RECURSIVE CASE
        return findSubstringRecursive(needle, haystack, i + 1)

print(findSubstringIterative('cat', 'My cat Zophie'))
print(findSubstringRecursive('cat', 'My cat Zophie'))
```

This *findSubstring.html* program has the equivalent JavaScript version:

JavaScript

```
<script type="text/javascript">
function findSubstringIterative(needle, haystack) {
    let i = 0;
    while (i < haystack.length) {
        if (haystack.substring(i, i + needle.length) == needle) {
            return i; // Needle found.
        }
        i = i + 1
    }
    return -1; // Needle not found.
}
```